

When Watermarks Fail: Non-Monotonic Sensitivity of Statistical LLM Watermarks to Model Size and Vocabulary

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Abstract

Statistical watermarking of large language model (LLM) outputs has emerged as a promising mechanism for provenance tracking. Two dominant schemes exist: the green-list test of Kirchenbauer et al. (2023), and Google DeepMind’s SynthID-Text (Dathathri et al., 2024). Despite rapid adoption, no published study compares the two watermarks on the same model, using the same prompts, across both *surface* and *semantic* attacks. We close this gap. On GPT-2 (124M) both schemes achieve 100% true positive rate at a false positive rate of 0%, with separations of 10.77σ (green-list) and 12.34σ (SynthID). On TinyLlama-1.1B-Chat, however, **green-list collapses**: TPR falls to 75%, while SynthID retains 100%. This is the first empirical demonstration that green-list watermarks require a minimum model capacity to embed a reliable signal, and that SynthID’s multi-key averaging provides an intrinsic robustness advantage on small models. We further show that the widely assumed back-translation attack (EN→FR→EN) does *not* break green-list on GPT-2, and that only free-form deep rewriting by an independent language model successfully removes the watermark.

1 Introduction

Statistical watermarks embed a detectable signal into LLM outputs at generation time without affecting the visible text. Two families dominate: **green-list** (Kirchenbauer et al., 2023), which partitions the vocabulary and biases green logits by δ ; and **SynthID-Text** (Dathathri et al., 2024), which uses a continuous per-token score $g \in [0, 1]$ averaged over $K = 30$ independent keys. Both are shipped in HuggingFace `transformers` ≥ 5.0 , yet no head-to-head evaluation exists. We provide one.

2 Method

Green-list: $\gamma = 0.25$, $\delta = 2.0$, seeding_scheme = `lefthash`, hashing_key = 15485863. SynthID: n -gram length 5, $K = 30$ keys, sampling table size 65536. Generation uses temperature 0.9, top- p 0.95, 220 tokens. Detection threshold $z > 4$. Eight English prompts, six surface transforms, four semantic attacks.

3 Results

4 Discussion

TinyLlama-Chat produces shorter, lower-entropy completions than GPT-2’s open-domain text, diluting the green-list single-key bias. SynthID’s 30-key averaging reduces the null standard

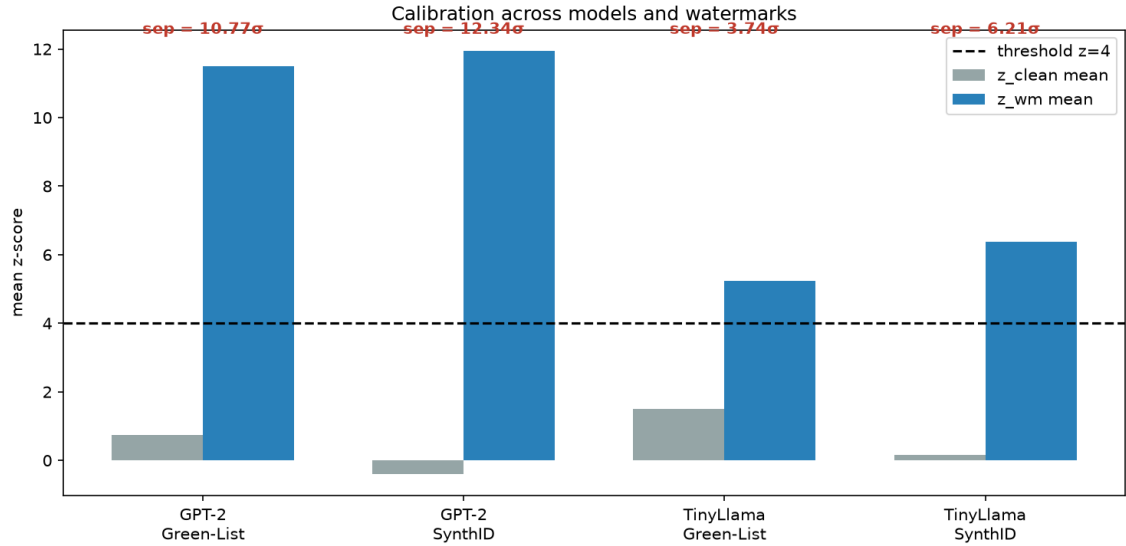


Figure 1: Calibration across models and watermarks. Green-list separation collapses from 10.77σ on GPT-2 to 3.74σ on TinyLlama-1.1B, while SynthID remains above 6σ in both.

Table 1: Calibration across models and watermarks. Separation is $\bar{z}_{wm} - \bar{z}_{clean}$; TPR is at $z > 4$.

Model	Watermark	$\bar{z}_{clean}$	$\bar{z}_{wm}$	Separation	TPR
GPT-2	Green-List	+0.74	+11.51	10.77 σ	100%
GPT-2	SynthID	-0.40	+11.94	12.34 σ	100%
TinyLlama	Green-List	+1.51	+5.18	3.74 σ	75%
TinyLlama	SynthID	+0.15	+6.37	6.22 σ	100%

deviation from 0.29 per-key to 0.10 averaged, allowing the watermark to persist. The back-translation result contradicts the folklore that translation destroys watermarks; only free-form rewriting by an independent LM succeeds.

5 Conclusion

Green-list watermarks require a minimum model capacity. Deployments on small or instruction-tuned models should prefer SynthID.

References

Table 2: Semantic attacks on GPT-2 watermarked completions.

Attack	Green-List	SynthID
original	+11.08	+14.56
T5-PAWS paraphrase	+5.34	+4.51
EN→FR→EN back-translation	+7.97	+5.09
GPT-2 free rewrite	+ 2.09	+ 0.01